

Geospatial Foundation Models Benchmarking

Datasets and Evaluation Challenges

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1 Current State of GFM Benchmarking Literature

Corley et al. [1] audited 152 papers related to Geospatial Foundation Models (GFMs) and argue that the current literature is substantially less comparable and reproducible than individual papers may suggest. Their study provides a systematic snapshot of the current GFM evaluation landscape.

The corpus also highlights some ambiguity in the use of the term “foundation model”. Only 46% of the 152 papers explicitly describe their proposed model as a foundation model in the title, abstract, or contributions. The remaining papers are earlier self-supervised remote-sensing models that previous GFM surveys already consider part of the same research lineage. The authors therefore treat the 46% figure as descriptive rather than as a criterion for inclusion.

The corpus was seeded from previous GFM surveys and expanded through two additional searches: an OpenAlex and Semantic Scholar citation-graph expansion, aimed mainly at recovering 2024–2025 papers not covered by the surveys, and a keyword search over remote-sensing self-supervised learning papers published between 2019 and 2023, before “foundation model” became common terminology in the field.

The resulting corpus contains 152 papers published between 2019 and 2025. Papers from 2026 were excluded because the year was incomplete at the time of submission. Metadata extraction was LLM-assisted and subsequently manually validated by the authors [1].

1.1 Model Weights Are Often Unavailable

Across the corpus, 39% of papers release no model weights [1]. A further 19% provide a public code repository but no released model artifact, meaning that reuse would require attempting to reproduce the original pretraining procedure.

This represents a basic obstacle to reproducible benchmarking. Before a model can be evaluated on a common downstream benchmark or used as a rerun baseline, the corresponding pretrained checkpoint must first be available.

1.2 The Field Does Not Converge on a Shared Core of Benchmarks

The 152 papers report evaluations on 401 distinct benchmarks, corresponding to 1,046 evaluation experiments in total [1].

The three most frequently used benchmarks—EuroSAT [2], NWPU-RESISC45 [3], and AID [4]—account for only 10.6% of all evaluations. The remaining 89.4% is distributed across 398 benchmarks, most of which appear in only one or two papers.

The problem is not simply that some benchmarks are more popular than others. Rather, individual papers frequently evaluate on almost disjoint benchmark sets.

For each paper, Corley et al. [1] compute the fraction of its downstream benchmarks that overlap with the ten most commonly used benchmarks in the corpus. The mean overlap is only 0.27, and 50 out of 143 papers, approximately 35%, have no overlap at all with the top ten.

At the corpus level, benchmark usage has a Gini coefficient of approximately 0.51. Moreover, this coefficient remains approximately stable after 2023, while the number of benchmarks appearing in only one paper increased from 13 in 2022 to 98 in 2025. The evidence therefore does not suggest convergence towards a shared evaluation core.

This creates a GFM-specific *benchmark lottery*: because models are rarely evaluated on sufficiently overlapping benchmark sets under common protocols, the published literature does not provide enough evidence to construct a reliable field-wide ranking of GFMs [1].

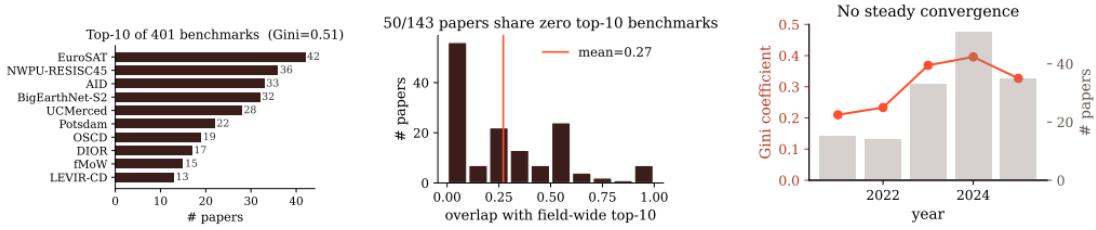


Figure 1: Benchmark usage across the audited GFM literature. The figure reports the most frequently used downstream benchmarks, per-paper overlap with the field’s top-10 benchmarks, and the evolution of benchmark concentration over time. Adapted from Corley et al. [1].

1.3 Reported Metrics Can Diverge Substantially Under the Same Nominal Protocol

The authors extracted every (model, benchmark, metric, evaluation protocol, training regime) tuple from the corpus, obtaining 10,817 results after benchmark and metric normalisation [1].

Evaluation protocols were grouped into categories such as fine-tuning, linear probing, kNN probing, zero-shot, and few-shot evaluation.

After filtering, 301 model–benchmark–protocol combinations appeared in at least two different papers. For each combination, the authors compute

$$\text{spread} = \max(\text{reported score}) - \min(\text{reported score}).$$

Of these 301 combinations, 76 have a spread of at least 5 percentage points, 46 at least 10 points, and 20 at least 20 points [1].

The largest discrepancy concerns Scale-MAE [5] evaluated on NWPU-RESISC45 using a linear probe. One paper reports an accuracy of 33.0, whereas the original Scale-MAE paper reports 89.6, corresponding to a difference of 56.6 percentage points, despite the use of the same released ViT-L checkpoint and the same nominal evaluation protocol [1].

This example illustrates that the label “linear probe” does not uniquely specify an experiment. Hidden choices may include the optimiser, learning rate, number of epochs, image resizing or cropping, normalisation, data augmentation, representation layer, train/validation/test split, class subset, sensor channels, and other adaptation details.

A second example reported by the authors concerns GPT-4o on UCMerced under zero-shot evaluation, where reported accuracy ranges from 43.5 to 88.8 percentage points. The corresponding papers do not describe the prompting configuration in sufficient detail to explain the discrepancy [1].

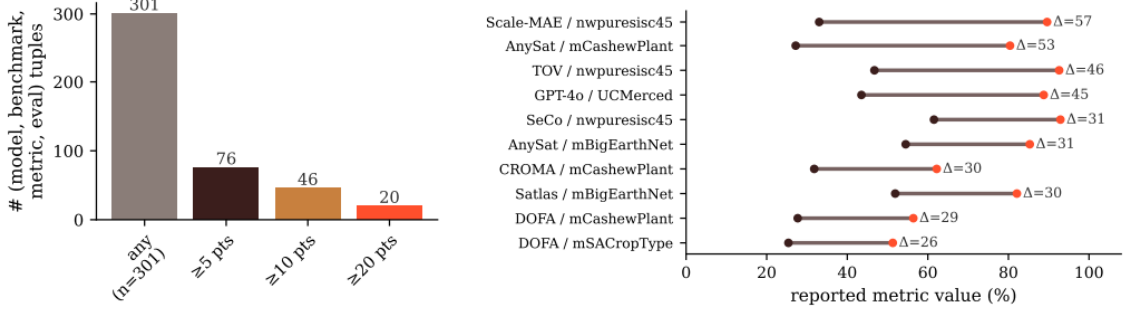


Figure 2: Largest cross-paper disagreements for repeated model–benchmark–protocol combinations. Even when the model, benchmark, and nominal evaluation protocol are the same, reported results can differ substantially. Adapted from Corley et al. [1].

Could This Be Explained by Random-Seed Variation?

The authors argue that random-seed variation alone is unlikely to explain the largest discrepancies.

Most repeated experiments agree closely, with a median spread close to zero, but the 90th-percentile spread is 12.7 percentage points [1].

Such a large tail is substantially greater than the run-to-run variation normally expected from simple classification heads under otherwise fixed conditions. The authors therefore interpret the largest disagreements primarily as evidence of hidden protocol differences.

1.4 Lack of a Shared Evaluation Harness

The geospatial community already provides benchmark collections and software infrastructure such as GEO-Bench [6], PANGAEA [7], and TorchGeo [8].

However, Corley et al. [1] argue that these resources do not yet constitute a community-wide evaluation harness comparable to those used in more mature evaluation ecosystems.

A benchmark collection may provide datasets, predefined splits, dataloaders, and example training scripts while still leaving substantial choices to the researcher. Consequently, two papers may both state that they evaluated on the same benchmark while actually executing different downstream experiments.

A stricter evaluation harness would additionally specify versioned task definitions, reference evaluation protocols, metrics, and automated checks, allowing a result to be reproduced from a model identifier and a task configuration.

Conceptually,

model → shared evaluation harness → standardised result.

Such a harness can reduce protocol-related degrees of freedom, but it cannot guarantee that the benchmark itself measures the intended deployment capability.

In particular, spatial dependence between training and test samples can inflate remote-sensing accuracy estimates [9, 10]. A perfectly reproducible evaluation may therefore still provide a misleading estimate of geographic generalisation.

Standardisation should also not necessarily imply forcing numerically identical hyperparameters on every architecture. Different models may require different optimisation settings. A fair comparison may instead require a common validation procedure and a comparable hyperparameter-search or compute budget, while allowing the final selected hyperparameters to differ between models.

1.5 Architecture and Pretraining Data Are Confounded

Another source of ambiguity arises when a new model simultaneously changes its architecture, pretraining data, and learning algorithm.

If several factors change at once, the observed downstream improvement cannot be attributed to one of them in isolation [1].

This is not an argument for requiring all GFMs to use identical pretraining data. Different, larger, private, or more diverse datasets may themselves be important contributions.

The problem arises when a paper attributes an improvement to an architectural or algorithmic modification without providing a controlled experiment in which the pretraining data are held fixed.

Corley et al. [11] show that apparent advantages of pretrained remote-sensing models over supervised ImageNet baselines can shrink substantially once preprocessing and distributional factors are controlled. The *No One Knows* study uses this as evidence that architecture and pretraining-data effects can be difficult to disentangle [1].

Similarly, Kaur et al. [12] study the effect of pretraining-data composition and show that changing the pretraining dataset alone can shift downstream performance by margins comparable to those often attributed to architectural novelty.

The corpus-level statistics make this comparability problem particularly clear.

Among the 126 papers for which a specific pretraining dataset could be extracted, the authors identify 87 distinct named primary pretraining datasets. Only 32 of the 126 papers, approximately 25%, share their complete pretraining configuration with at least one other paper. The remaining 94 use a configuration not shared by any other work in the corpus [1].

Consequently, comparisons in the GFM literature often involve systems that differ simultaneously in architecture, pretraining data, training objective, and downstream evaluation protocol.

Such comparisons remain useful at the *system level*, but they cannot by themselves establish which individual design choice caused a performance improvement.

1.6 Recommendations

Corley et al. [1] propose six recommendations aimed at improving comparability across the field:

1. **Release reusable model weights**, or explicitly state why they cannot be released.
2. **Evaluate on a minimum shared core of benchmark tasks**, while adding task-specific evaluations when needed.
3. **Distinguish copied and independently rerun baseline results**, including the corresponding source or configuration.
4. **Report uncertainty or repeated-run variability** when computationally feasible.
5. **Develop a shared third-party evaluation harness** with versioned protocols, common task definitions, and automated checks.
6. **Disentangle pretraining data, architecture, and learning algorithm** through controlled comparisons when causal claims are made.

2 Benchmark Datasets

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